GinCol Lab · Aula

Student worksheet

Biology, Geology and Environmental Sciences · 1st year of Bachillerato · Inside the Earth · Class 3 of 5 · With Nil

The shadow of the core

Class video: https://youtu.be/v9dhRXUVfVk

The code leads to the video.

Name: _____ Date: _____

Before watching

1. A large earthquake breaks the rock 600 km beneath the Fiji Islands, in the Pacific. Do you think seismometers in Spain will record it? How long will the waves take to arrive? Write down your answer.

2. In the previous class you saw the deepest borehole ever drilled. Where is it, and how deep is it?

3. **Your prediction.** If a wave travels through the inside of the Earth to stations farther and farther away, do you think it will reach all of them? Why?

While you watch the video

1. What is an indirect method? Give an example.

2. What does a seismometer record? Why does the mass almost stay still when the ground moves?

3. How does a P wave move the rock? And an S wave? Which one arrives first?

4. Why don't S waves go through liquids?

5. **Your prediction.** In the Hawaii record, which wave will arrive first, and in which of the two records, the up-and-down one or the sideways one, will it show up better? Why?

6. How long did the P wave take to reach Hawaii? And the S, after the P? What happens to that gap the farther away the station is?

7. What are surface waves? How are they different from the P and the S: where do they travel, how do they move the ground and when do they arrive?

8. Why was a deep earthquake chosen for this class, and not one near the surface?

9. What does it mean that a station is 60° from the earthquake? Roughly how many kilometres is that?

10. **Your bet.** If we place each record at the height of its distance, how far will the P wave reach? A: everywhere, equally strong. B: everywhere, but weaker and weaker. C: up to a certain distance, and there it fades out.

11. In the record section of the 96 records, what do you see? What catches your attention?

12. How many times weaker does the P arrive at 90° than at 30°? And at 120° than at 90°? Why do we need this graph, if we already have the record section?

13. What is a control? How does the Toledo station show that it is not broken?

14. Between what distances is the shadow ring? Which cities were inside it that day? And Toledo?

15. Your textbook may say that the shadow runs from about 103° to about 143°. Why, with these data, does it start at around 100°, or earlier?

16. **Stop and think.** What could cast a shadow like this, the same in all directions and with any earthquake?

17. In this class, what is a fact and what is an interpretation?

18. Is it true that nothing reaches the shadow zone? What does arrive?

19. Write the two puzzles from the end in your own words. Can you think of an explanation?

In class: the shadow, hands-on

You will work with the printed record section and with this table, of 29 of the 96 stations that recorded the Fiji earthquake of 19/08/2018. The **arrival** is the minute, after the earthquake, when the ground starts to move clearly. The **strength** is the maximum speed of the ground in that arrival, in microns (thousandths of a millimetre) per second, µm/s. A dash, "—", means that wave cannot be found in the record.

Station	Where	Distance (°)	P arrival (min)	What arrives at the time of the P (µm/s)	Core wave arrival (min)	Core wave strength (µm/s)
MSVF	Monasavu (Fiji)	3.6	1.22	3,919	—	—
RAR	Rarotonga (Cook Islands)	17.6	3.46	222	—	—
EIDS	Eidsvold (Australia)	29.4	5.21	128	—	—
COEN	Coen (Australia)	37.3	6.34	307	—	—
WRAB	Tennant Creek (Australia)	44.8	7.30	276	—	—
KNRA	Kununurra (Australia)	50.7	8.05	469	—	—
GIRL	Giralia (Australia)	63.0	9.49	292	—	—
MAJO	Matsushiro (Japan)	68.4	9.97	57	—	—
YSS	Yuzhno-Sakhalinsk (Russia)	74.0	10.48	66	—	—
TUC	Tucson (United States)	81.8	11.24	54	—	—

Station	Where	Distance (°)	P arrival (min)	What arrives at the time of the P (µm/s)	Core wave arrival (min)	Core wave strength (µm/s)
ANMO	Albuquerque (United States)	86.2	11.60	25	—	—
PLCA	Paso Flores (Argentina)	90.9	12.01	19	—	—
ULN	Ulaanbaatar (Mongolia)	93.6	12.15	30	—	—
LCO	Las Campanas (Chile)	95.7	12.42	8.8	—	—
CCM	Cathedral Cave (United States)	98.7	—	3.8	—	—
WVT	Waverly (United States)	100.8	—	2.2	—	—
LPAZ	La Paz (Bolivia)	103.0	—	2.2	—	—
CPUP	Villa Florida (Paraguay)	107.4	—	0.83	—	—
KURK	Kurchatov (Kazakhstan)	112.2	—	0.81	—	—
KBL	Kabul (Afghanistan)	118.7	—	0.24	17.80	24
ARU	Arti (Russia)	123.2	—	0.21	17.83	38
BOSA	Boshof (South Africa)	128.1	—	0.07	18.05	14
OBN	Obninsk (Russia)	134.6	—	0.11	18.14	20

Station	Where	Distance (°)	P arrival (min)	What arrives at the time of the P (µm/s)	Core wave arrival (min)	Core wave strength (µm/s)
KMBO	Kilima Mbogo (Kenya)	140.2	—	0.02	18.35	25
SHEL	Saint Helena Island (South Atlantic)	145.1	—	0.77	18.59	177
GRFO	Gräfenberg (Germany)	147.5	—	0.06	18.59	132
SSB	Saint-Sauveur-en-Rue (France)	152.7	—	0.03	18.63	25
CLTB	Caltabellotta (Italy)	158.1	—	—	18.82	58
TAM	Tamanrasset (Algeria)	174.2	—	—	19.02	12

Data: NSF NGF archive of EarthScope Consortium, open global networks, CC BY 4.0.

1. **Your prediction.** Before looking at the data: how far will the P wave reach? A, B or C?

2. **By eye, on the record section.** With a ruler, on six records (near 10°, 30°, 60°, 90°, 110° and 150°), mark where the line starts to move and read the minute. Which ones are hard? Remember: each record is enlarged to fit, so the size of the waves cannot be compared.

3. **The graph.** On graph paper, put the minutes on the horizontal axis (from 0 to 20, one centimetre per minute) and the distance on the vertical axis (from 0° to 180°, one centimetre for every 10°), as in the record section. With one colour, the arrival of the P at each station; with another, the arrival of the core wave.
    - Where does the P curve end?
    - Where does the other branch begin? How many minutes later?
    - Is the P curve a straight line?
4. **The strength.** In the column "what arrives at the time of the P":
    - From what distance does it drop below 1 µm/s?
    - How many times weaker is it at ARU (123.2°) than at ULN (93.6°)?
    - Where does the core wave arrive most strongly?

5. **Conclude.**
    - Between what distances is the shadow ring?

    - The Kabul station does not record the direct P wave. Is it broken? How do you know?

    - Does what you have found match what your textbook says? If not, why might that be?

Practice

1. The focus of the Fiji earthquake was 600 km deep. How many times deeper is that than the deepest borehole ever drilled (12,262 m)?

2. In the record section, how long does the P wave take to reach a station at 60°?

3. Why is a station that does not record the P from an earthquake not necessarily broken?

Try it yourself

With a toy spring stretched out on the floor or on a long table, and someone holding the other end:

1. Push your end back and forth, along the spring. How do the coils move? That is a P wave.

2. Now shake it from side to side. How does the spring move this time? That is an S wave.

3. What does the spring show well about earthquake waves? What can't it show?

Keep thinking

Seismology was born to study earthquakes and ended up discovering the centre of the planet. Is that normal in science, finding what you weren't looking for?

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